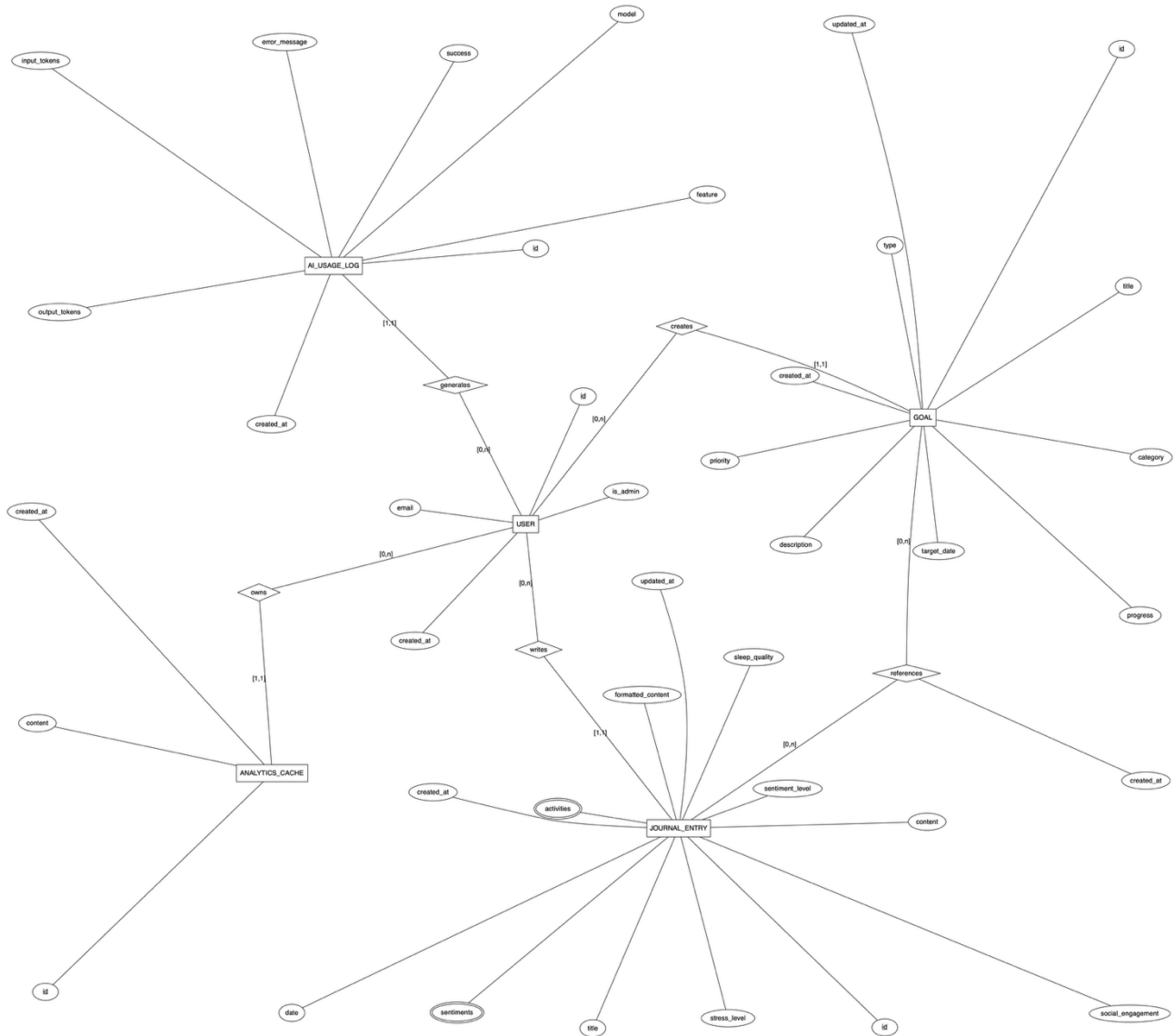


Reflecta

ER-Diagram



One-Table-Design

ENTITY	PK	SK	ATTRIBUTES
User	USER	<sub>	email, is_admin, created_at
JournalEntry	USER#<sub>	ENTRY#<entryId>	title, date, content, sentiment_level, sleep_quality, stress_level,

			social_engagement, formatted_content, activities, sentiments, created_at, updated_at
Goal	USER#<sub>	GOAL#<goalId>	title, type, target_date, category, priority, description, progress, created_at, updated_at
Entry→Goal Link	USER#<sub>	ENTRY#<entryId> #GOAL#<goalId>	created_at
Goal→Entry Link	USER#<sub>	GOAL#<goalId>#ENTRY#<entryId>	created_at
AnalyticsCache	USER#<sub>	CACHE#<cache_type> #<period>	content, created_at
AIUsageLog	USER#<sub>	AILOG#<created_at> #<logId>	feature, model, input_tokens, output_tokens, success, error_message, created_at

Local Secondary Indices

LSI1 — Entries by date (per user)

ATTRIBUTE	VALUE
pk	USER#<sub>
LSI1_sk	ENTRY#DATE#<date>

LSI2 — Goals by category (per user)

ATTRIBUTE	VALUE
pk	USER#<sub>
LSI2_sk	GOAL#CAT#<category>

LSI3 — Goals by priority (per user)

ATTRIBUTE	VALUE
pk	USER#<sub>
LSI3_sk	GOAL#PRIO#<priority>

Global Secondary Indices

GSI1 — User by email (User Login)

ATTRIBUTE	VALUE
GSI1_pk	EMAIL#<email>
GSI1_sk	USER#<sub>

GSI2 — Admin / cross-user lookups

ATTRIBUTE	VALUE
GSI2_pk	ENTITY#<type>
GSI2_sk	<created_at>#<id>

GSI3 — AI usage analytics (cross-user)

ATTRIBUTE	VALUE
GSI3_pk	AILOG#<feature>
GSI3_sk	<created_at>#<logId>

Access Patterns

User

ACCESS PATTERN	QUERY
Get user profile by sub	pk = USER, sk = <sub>
Get user by email (login)	GSI1_pk = EMAIL#<email>
List all users	pk = USER
Admin: List users created in date range	GSI2_pk = ENTITY#USER, GSI2_sk BETWEEN <t1> AND <t2>

JournalEntry

ACCESS PATTERN	QUERY
Get one journal entry	pk = USER#<sub>, sk = ENTRY#<entryId>
List all entries of a user	pk = USER#<sub>, begins_with(sk, "ENTRY#")
List entries of a user, sorted by date	pk = USER#<sub>, begins_with(LSI1_sk, "ENTRY#DATE#")
List entries of a user in a date range	pk = USER#<sub>, LSI1_sk BETWEEN ENTRY#DATE#<d1> AND ENTRY#DATE#<d2>
Admin: List all entries	GSI2_pk = ENTITY#ENTRY
Admin: List entries created in date range	GSI2_pk = ENTITY#ENTRY, GSI2_sk BETWEEN <t1> AND <t2>

Goal

ACCESS PATTERN	QUERY
Get one goal	pk = USER#<sub>, sk = GOAL#<goalId>
List all goals of a user	pk = USER#<sub>, begins_with(sk, "GOAL#")
List goals of a user by category	pk = USER#<sub>, LSI2_sk = GOAL#CAT#<category>
List all categorized goals (sorted by category)	pk = USER#<sub>, begins_with(LSI2_sk, "GOAL#CAT#")
List goals of a user by priority	pk = USER#<sub>, LSI3_sk = GOAL#PRIO#<priority>
List all goals sorted by priority	pk = USER#<sub>, begins_with(LSI3_sk, "GOAL#PRIO#")
Admin: List all goals	GS12_pk = ENTITY#GOAL
Admin: List goals created in date range	GS12_pk = ENTITY#GOAL, GS12_sk BETWEEN <t1> AND <t2>

Entry ↔ Goal Links

ACCESS PATTERN	QUERY
List goals linked to one entry	pk = USER#<sub>, begins_with(sk, "ENTRY#<entryId>#GOAL#")
List entries linked to one goal	pk = USER#<sub>, begins_with(sk, "GOAL#<goalId>#ENTRY#")
Check if specific entry/goal pair is linked	pk = USER#<sub>, sk = ENTRY#<entryId>#GOAL#<goalId>

AnalyticsCache

ACCESS PATTERN	QUERY
Get cached analytics for a user (type + period)	pk = USER#<sub>, sk = CACHE#<cache_type>#<period>
List all caches of a user	pk = USER#<sub>, begins_with(sk, "CACHE#")
List all caches of a specific type for a user	pk = USER#<sub>, begins_with(sk, "CACHE#<cache_type>#")

AIUsageLog

ACCESS PATTERN	QUERY
List all AI usage logs of a user	pk = USER#<sub>, begins_with(sk, "AILOG#")
List AI usage logs of a user in time range	pk = USER#<sub>, sk BETWEEN AILOG#<t1> AND AILOG#<t2>

Admin: List AI usage logs by feature (cross-user)	GSI3_pk = AILog#<feature>
Admin: List AI usage logs by feature in time range (cross-user)	GSI3_pk = AILog#<feature>, GSI3_sk BETWEEN <t1> AND <t2>